5.3.9 Internal clock timing characteristics

The parameters given in the tables below are derived from tests performed under ambient temperature and supply voltage conditions summarized in Table 16. The curves provided are characterization results, not tested in production.

5.3.9.1 *High-speed internal HSI144 oscillator*

Table 35. HSI144 oscillator characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f_{HSI}(1)	HSI frequency	V_{DD} = 3.3 V, T_J = 30 °C	144.07	-	145.08	MHz
TRIM(2)	USER trimming step	-	-	0.1	0.15	%
USER TRIM COVERAGE(2)	USER TRIMMING Coverage positive	80 steps	5.2%	8%	-	%
	USER TRIMMING Coverage negative	48 steps	-3.1%	-4.8%	-	%
DuCy(HSI)(2)	Duty Cycle	-	45		55	%
Δ_{TEMP} (HSI)(3)	HSI oscillator frequency drift over temperature (the reference is 144 MHz.)	T_J = -20 to 130 °C	-1	-	1	%
		T_J = −40 to T_{Jmax} °C	-1.5	-	1.5	%
$\Delta_{VDD(HSI)}$(2)(4)	HSI oscillator frequency drift with V_{DD}Section 5.3.9.1: High-speed internal HSI144 oscillator Section 5.3.9.1: High-speed internal HSI144 oscillator(the reference is 3.3V)	V_{DD} from 2.7 V to 3.6 V	-	-	- 0.1	%
$t_{su(HSI)}$(3)	HSI oscillator start-up time (PSI Off)	-	-	3	4.5	µs
	HSI oscillator start-up time (PSI On)	-	-	0.5	-	µs
t_{stab}(2)	stabilization time (PSI OFF from Enable)	+/-1% of target freq	-	-	8	µs
	stabilization time (PSI ON from Enable)		-	-	0.7	µs
$I_{DD(HSI)}$(2)(5)	HSI supply regulation block oscillator power consumption	-	-	91	-	µA
	HSI oscillator power consumption	-	-	28	-	µA
N_T jitter(2)	Next transition jitter(6).	On HSIDIV3	-	52	322	ps
P_T jitter(2)	Paired transition jitter(7)		-	62	394	ps
P_{er} jitter(2)	Period Jitter standard deviation	On HSIS		15		ps
		On HSIDIV3		26		ps

1. *Tested in production.*
2. *Specified by design - Not tested in production.*
3. *Evaluated by characterization - Not tested in production.*
4. $\Delta f_{HSI} = \Delta TEMP + \Delta_{VDD}$
5. *The supply regulation consumption is common to HSI and PSI. (To be counted once if both oscillators are ON).*
6. *Jitter measurements are performed without clock source activated in parallel. Typical value is standard deviation, Maximum is peak measure on TIE-8 over 36 cycles.*
7. *Jitter measurements are performed without clock source activated in parallel. Typical value is standard deviation, Maximum is peak measure on TIE-16 over 36 cycles.*